

When Inverse Physics-Informed Neural Networks Become Practically Identifiable: A Case Study in Synaptic Dopamine Transport

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Abstract

Quantifying dopamine clearance matters for understanding dopaminergic dysfunction in Parkinson’s disease, but classical mesh-based methods struggles under sparse, noisy observation. We formulate recovery of the diffusion coefficient $\mathbf{D}$ and reuptake rate $\mathbf{k}$ as a parameter inverse problem for a 2D reaction-diffusion PDE of synaptic dopamine transport, using physics-informed neural networks (PINNs) as a mesh-free forward and inverse solver. All validation uses synthetic data from a finite-difference reference; no biological measurements are used yet. The forward model attains an L2 relative error of **1.52%** against the finite-difference reference and **3.75%** against the analytical solution. For the inverse problem, evaluated over a five-seed ensemble, we recover $\mathbf{D} = \mathbf{0.3515} \pm \mathbf{0.0034} \mu\text{m}^2/\text{ms}$ and $\mathbf{k} = \mathbf{0.0573} \pm \mathbf{0.0006} \text{ms}^{-1}$, with mean relative errors of **9.85%** and **14.61%**. The inverse problem is well-conditioned along one direction and ill-conditioned along a diffusion-reuptake trade ray, with implications for real-data experimental design. Unlike a Levenberg-Marquardt baseline, which drives $\mathbf{k}$ to non-physical negative values under bounded-domain data, the PINN’s residual-based formulation recovers physically meaningful parameters. This identifiability-aware framework is a reusable tool for inverse-PDE recovery of synaptic neurotransmitter dynamics; application to experimental fast-scan cyclic voltammetry data is the immediate next step.

Keywords: Physics-informed neural networks, dopamine diffusion, reaction-diffusion equations, inverse problem, Parkinson’s disease

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Glossary of Terms

$C(x, y, t)$	dopamine concentration field	μM
D	effective diffusion coefficient	$\mu\text{m}^2/\text{ms}$
k	linearised dopamine transporter (DAT) reuptake rate constant	ms^{-1}
L	domain side length	μm
T	simulation time window	ms
σ	release-pulse half-width	μm
C_0	peak release concentration	μM

1 Introduction

Parkinson’s disease (PD) is the second most prevalent neurodegenerative disorder worldwide, affecting approximately 10 million people and imposing a substantial public-health burden [9]. Its hallmark pathology is progressive degeneration of dopaminergic neurons in the substantia nigra pars compacta, reducing dopamine availability in the striatum and producing the cardinal motor symptoms of PD: bradykinesia, rigidity, and tremor [7, 13].

Dopaminergic signaling begins when an action potential triggers Ca^{2+} -dependent vesicle fusion, releasing $\sim 10^3$ – 10^4 dopamine molecules into the ~ 20 nm synaptic cleft [25], which then diffuses with effective coefficient $D \approx 0.3 \mu\text{m}^2/\text{ms}$ [5, 17] and is cleared by the dopamine transporter (DAT), modeled at sub-saturating concentrations as a first-order reuptake rate k [5]. PD reduces both functional release-site density and striatal DAT expression [5, 20], so changes in (D, k) are a quantitatively meaningful proxy for disease state. Fast-scan cyclic voltammetry (FSCV) provides millisecond-resolution, sub-micrometer dopamine recordings *in vivo* and *ex vivo* [21], but yields exactly the sparse, noisy regime in which recovering (D, k) is hardest.

Classical reaction-diffusion models of dopamine transport [17, 20, 25] use finite-difference or finite-element solvers, requiring costly mesh construction in complex neuroanatomical geometries and repeated forward solves per optimization step, an approach that scales poorly and is ill-suited to sparse experimental data. Physics-informed neural networks (PINNs) [19] instead embed the governing PDE directly into the network’s training objective, solving forward and inverse problems mesh-free.

PINNs have been applied to cardiac electrophysiology [23], systems-biology kinetics [26], and PDE discovery from neural imaging [22], but, to our knowledge and conditional on the limits of our search, not to synaptic neurotransmitter diffusion.

This paper makes four contributions. First, an inverse-PINN formulation for spatially-resolved 2D dopamine transport, extending prior inverse-PINN work in systems biology [26], lumped ODE kinetics without spatial structure to a spatially explicit problem. Second, forward validation against both an analytical solution and a finite-difference reference, with relative L^2 errors comparable to established numerical schemes without a spatial mesh. Third, the inverse problem recovering D and k from sparse, noisy synthetic data, characterizing its practical identifiability: well-conditioned along one direction, ill-conditioned along a diffusion-reuptake trade ray, with direct implications for experimental design; we further show the PINN’s bounded-domain residual formulation succeeds where a classical Levenberg-Marquardt baseline fails, driving k negative under the same data. Fourth, we release the complete implementation seed-controlled scripts and a reproducible notebook as an open-source, ready-to-extend baseline for application to experimental FSCV data, the immediate next step toward a patient-specific tool for dopaminergic dysfunction in PD.

2 Methods

2.1 Model Formulation

We model dopamine concentration $C(x, y, t)$ on a 2D domain spanning the synaptic cleft and surrounding perisynaptic space. The 2D restriction is motivated by the cleft’s thinness ($h \approx 20\text{--}30\text{ nm}$, two orders of magnitude below the lateral spreading scale $\sqrt{2DT} \approx 3.6\text{ }\mu\text{m}$, so vertical equilibration is essentially instantaneous), by consistency with prior striatal dopamine modeling [25], and because it admits a closed-form analytical reference. Extension to 3D is a straightforward modification of the input layer and PDE residual (Section 4.5).

Following Cragg and Rice [5], DAT action at sub-saturating concentrations is approximated by a first-order linear sink $-kC$ (accurate when $C \ll K_m \approx 0.2\text{ }\mu\text{M}$), reducing recovery to two scalars (D, k). Release is modeled as a radially symmetric Gaussian pulse at $t = 0$, consistent with the sub-millisecond timescale of vesicular exocytosis [25].

With these modeling choices, we consider the reaction-diffusion initial-boundary-value problem on the square domain $\Omega = (-L/2, L/2)^2$, $t \in (0, T]$,

$$\frac{\partial C}{\partial t} = D \nabla^2 C - kC, \quad (x, y) \in \Omega, \quad t \in (0, T], \quad (1)$$

where $\nabla^2 C = \partial_{xx}C + \partial_{yy}C$ is the two-dimensional Laplacian, with the radially symmetric Gaussian initial condition

$$C(x, y, 0) = C_0 \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right), \quad (x, y) \in \bar{\Omega}, \quad (2)$$

Table 1 Model parameters for the dopamine synaptic diffusion problem. D , k from canonical experimental values for striatal tissue; L , σ , C_0 from the Wiencke et al. [25] model; T is a modelling choice.

Parameter	Symbol	Value	Source
Diffusion coefficient	D	$0.32 \mu\text{m}^2/\text{ms}$	[5, 17]
Reuptake rate constant	k	0.05 ms^{-1}	[5]
Domain length	L	$5.0 \mu\text{m}$	[25]
Simulation time	T	20 ms	(model choice)
Peak source amplitude	C_0	$1.0 \mu\text{M}$	[25]

and zero-flux (Neumann) boundary conditions on all four edges of $\partial\Omega$,

$$\left. \frac{\partial C}{\partial x} \right|_{x=\pm L/2} = 0, \quad y \in [-L/2, L/2], \quad t \in (0, T], \quad (3)$$

$$\left. \frac{\partial C}{\partial y} \right|_{y=\pm L/2} = 0, \quad x \in [-L/2, L/2], \quad t \in (0, T]. \quad (4)$$

The zero-flux conditions (3)–(4) encode chemical isolation from neighboring synaptic compartments on the timescales of interest. The Gaussian profile in (2) represents the instantaneous footprint of a single quantal release event, with σ the release radius and C_0 the peak concentration.

All model parameters are taken from the experimental dopamine literature for striatal tissue [5, 17] and are summarized in Table 1.

2.2 Implementation and Reproducibility

Detailed JAX/Flax NNX code listings for the network module, PDE residual, and inverse-problem training loop are provided in Online Resource 1; the closed-form analytical solution derivation underlying the forward-problem validation reference is provided in Online Resource 2.

2.3 PINN Architecture

We employ a fully-connected feedforward network $\mathcal{N}_\theta : \mathbb{R}^3 \rightarrow \mathbb{R}$ mapping space-time coordinates (x, y, t) to predicted concentration $\hat{C}(x, y, t; \theta)$, with four hidden layers of 64 neurons each and tanh activation (continuously differentiable, the standard PINN choice [19]):

$$\hat{C} = \mathcal{N}_\theta(x, y, t) = (W_\ell \circ \sigma \circ W_{\ell-1} \circ \cdots \circ \sigma \circ W_1)(x, y, t), \quad (5)$$

with $\ell = 5$ weight matrices (four hidden layers plus a linear output layer, allowing $\hat{C}$ unbounded range). Weights use Glorot initialization [10]; the network has 12,801 trainable parameters in total.

Spatial and temporal derivatives in the PDE residual (1) are evaluated by automatic differentiation of $\mathcal{N}_\theta$, removing any need for finite-difference stencils. The

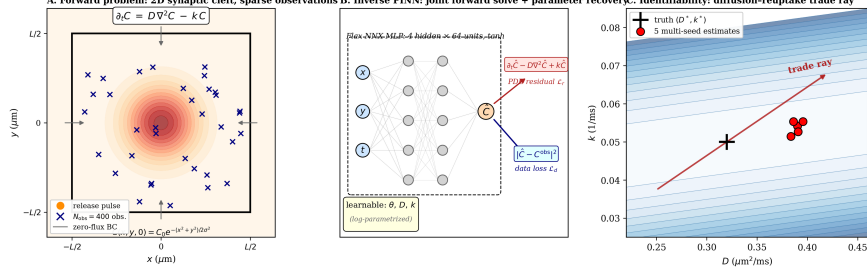


Fig. 1 Overview of the inverse-PDE problem and the physics-informed neural network methodology. **A.** Forward setting: 2D reaction-diffusion PDE on $\Omega = [-L/2, L/2]^2$ with Neumann boundaries and a Gaussian release pulse; crosses mark a subset of the $N_{\text{obs}} = 400$ sparse noisy observations. **B.** Inverse PINN: a Flax NNX MLP $\mathcal{N}_\theta : (x, y, t) \mapsto \hat{C}$ is trained to satisfy the PDE residual while fitting sparse noisy observations; D and k are log-parametrized learnable scalars optimized jointly with the network weights. **C.** Identifiability: a one-dimensional “trade ray” through (D, k) space along which the loss is flat to within the noise scale. Five seeds (red dots) cluster along this ridge above ground truth $(D^*, k^*) = (0.32, 0.05)$, reflecting the structural limit of Section 4.3

implementation uses JAX [2] with the network as a Flax NNX module [12], Adam via Optax [6], and L-BFGS via jaxopt [1]; JIT compilation gives a 2–5 $\times$ speed-up over a comparable PyTorch implementation. Complete source code and seed-controlled training scripts are publicly available (Declarations: Code availability).

2.4 Loss Function Formulation

The total PINN loss function is the weighted sum of four terms:

$$\mathcal{L}(\theta) = \lambda_r \mathcal{L}_r + \lambda_i \mathcal{L}_i + \lambda_b \mathcal{L}_b + \lambda_d \mathcal{L}_d, \quad (6)$$

The scalar weights $\lambda_r, \lambda_i, \lambda_b, \lambda_d \geq 0$ control the relative contribution of each term; imbalanced weights induce gradient pathologies that prevent convergence [24]. The individual terms are defined as follows.

Residual loss $\mathcal{L}_r$

enforces the PDE at N_r collocation points $\{(x_j^r, y_j^r, t_j^r)\}_{j=1}^{N_r}$:

$$\mathcal{L}_r = \frac{1}{N_r} \sum_{j=1}^{N_r} \left| \frac{\partial \hat{C}}{\partial t} \Big|_j - D \nabla^2 \hat{C} \Big|_j + k \hat{C}_j \right|^2. \quad (7)$$

Initial condition loss $\mathcal{L}_i$

at N_i points:

$$\mathcal{L}_i = \frac{1}{N_i} \sum_{j=1}^{N_i} \left| \hat{C}(x_j^i, y_j^i, 0) - C_0(x_j^i, y_j^i) \right|^2. \quad (8)$$

Boundary condition loss $\mathcal{L}_b$

at N_b boundary points:

$$\mathcal{L}_b = \frac{1}{N_b} \sum_{j=1}^{N_b} \left| \mathcal{B}[\hat{C}](x_j^b, y_j^b, t_j^b) \right|^2, \quad (9)$$

where $\mathcal{B}[\cdot]$ denotes the boundary condition operator corresponding to (3)–(4).

Data loss $\mathcal{L}_d$

(for inverse problem; set $\lambda_d = 0$ for forward problem) at N_d observation points:

$$\mathcal{L}_d = \frac{1}{N_d} \sum_{j=1}^{N_d} \left| \hat{C}(x_j^d, y_j^d, t_j^d) - C_j^{\text{obs}} \right|^2. \quad (10)$$

We upweight the residual, using $(\lambda_r, \lambda_i, \lambda_b) = (10, 1, 1)$ for the forward problem ($\lambda_d = 0$) and $(10, 1, 1, 1)$ for the inverse problem, keeping the residual on the same gradient scale as the other terms and preventing the optimizer from fitting observations at the expense of physical consistency. Weights are fixed throughout; adaptive schemes [24] are future work.

2.5 Collocation Point Sampling

Collocation points are sampled from $\Omega \times (0, T]$ using Latin Hypercube Sampling, which provides better coverage than uniform random sampling [16]: $N_r = 10,000$ interior points, $N_i = 400$ initial-condition points, and $N_b = 400$ boundary points (100 per edge), plus 5,000 held-out test points for monitoring generalization. All samples regenerate from a fixed seed for reproducibility.

2.6 Training Procedure

Training proceeds in two phases: Adam [14] for 20,000 iterations at fixed learning rate 10^{-3} , followed by limited-memory BFGS [15] via jaxopt [1] to convergence (max. 5,000 iterations, gradient tolerance 10^{-9}), exploiting second-order curvature once iterates enter a local minimum’s neighborhood. Loss weights follow Section 2.4 and are held fixed throughout; adaptive reweighting schemes [24] are left for future work.

All experiments use a fixed seed (1234) for floating-point reproducibility. Forward-problem training completes in 3–5 minutes on a T4 GPU (10–15 on CPU); the full reproduction pipeline (forward training, validation, five-seed inverse ensemble, figure generation) completes in ~ 50 minutes on a T4 GPU (Online Resource 1 for detail).

2.7 Inverse Problem Formulation

The forward problem of Section 2.1 assumes D, k are known and seeks $C(x, t)$; the inverse setting reverses this, recovering (D, k) from a partial, noisy observation. Classical mesh-based solvers are expensive here, requiring a separate nested forward solve

per candidate (D, k) , with cost growing linearly in candidates [25]. PINNs sidestep this by treating D and k as additional trainable network parameters, so one training run recovers both the field and the physics constants [11, 19].

Synthetic observations $\{(x_j^d, y_j^d, t_j^d, C_j^{\text{obs}})\}_{j=1}^{N_d}$ are generated by sampling locations uniformly from $\Omega \times [t_{\min}, t_{\max}]$ with $t_{\min} = 0.5 \text{ ms}$ (just past the initial-condition transient, so data carry information about both D and k) and $t_{\max} = 0.9 T = 18 \text{ ms}$ (the window over which the pulse remains well-confined; cf. Online Resource 2). At each location, the true concentration is computed by interpolating the finite-difference reference solver (at $D = D^*$, $k = k^*$) and corrupted by additive Gaussian noise,

$$C_j^{\text{obs}} = C^*(x_j^d, y_j^d, t_j^d) + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, (\eta C_0)^2), \quad (11)$$

with η the noise level as a fraction of C_0 . Using the finite-difference (rather than analytical) reference keeps the observations exactly consistent with the Neumann boundary conditions enforced by the PINN, avoiding a data-model mismatch that would otherwise bias recovery at large t . Unless stated otherwise, $N_d = 400$ and $\eta = 0.02$ (2%, comparable to high-quality FSCV noise floors [21]).

D and k are introduced as trainable variables in the same back-propagation graph as θ , parametrized in log space ($D = \exp(\tilde{D})$, $k = \exp(\tilde{k})$) to enforce positivity. Writing $\Phi = (\theta, \tilde{D}, \tilde{k})$, the inverse PINN minimizes

$$\Phi^* = \arg \min_{\Phi} [\lambda_r \mathcal{L}_r(\Phi) + \lambda_i \mathcal{L}_i(\Phi) + \lambda_b \mathcal{L}_b(\Phi) + \lambda_d \mathcal{L}_d(\Phi)], \quad (12)$$

with $(\lambda_r, \lambda_i, \lambda_b, \lambda_d) = (10, 1, 1, 1)$, so the optimizer prioritizes physical consistency over fitting observation noise. Trainable parameters are initialized perturbed from truth ($\exp(\tilde{D}_0) = 0.30$, $\exp(\tilde{k}_0) = 0.04$); the optimization schedule matches the forward problem (Section 2.6).

This is motivated by Section 1: both parameters are expected to shift in Parkinson’s disease, and FSCV delivers exactly the sparse, noisy measurements the inverse PINN consumes [21]. This study restricts itself to controlled synthetic validation; extending to real FSCV recordings is a required next step, discussed in Section 4.

3 Results

3.1 Forward Problem: Accuracy and Convergence

Fig 2 compares the trained PINN prediction $\hat{C}(x, t; \theta)$ with the closed-form analytical solution and the explicit finite-difference reference at three representative time snapshots ($t = 0$, $T/4 = 5 \text{ ms}$, $T/2 = 10 \text{ ms}$). The three curves coincide to within line thickness across the entire spatial domain at every snapshot. The full space-time error field (Fig 3) shows the pointwise absolute error $|C - \hat{C}|$ remains several orders of magnitude below C_0 across the entire evaluation grid.

The PINN attains an L2 relative error of 1.52% against a finite-difference (FD) reference, integrated across the full $201 \times 201 \times 200$ space-time grid, and 3.75% against the closed-form infinite-domain analytical solution (Online Resource 2) at $t = 1 \text{ ms}$.

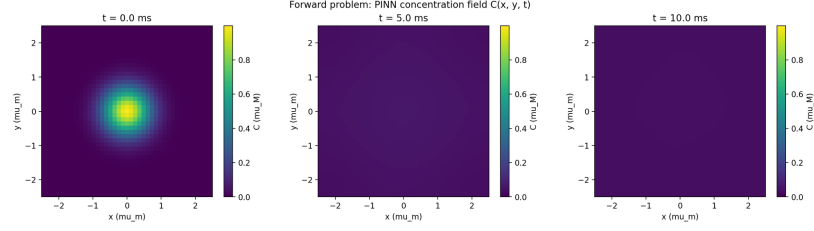


Fig. 2 Forward-problem solution: PINN vs. analytical and finite-difference references. $C(x, t)$ at $t = 0$, $T/4 = 5$ ms, and $T/2 = 10$ ms. Analytical solution (solid black), finite-difference reference (dotted blue), and PINN prediction $\hat{C}$ (dashed red) are visually indistinguishable at all three slices. Parameters: $D = 0.32 \mu\text{m}^2/\text{ms}$, $k = 0.05 \text{ ms}^{-1}$, $L = 5 \mu\text{m}$, $\sigma = 0.5 \mu\text{m}$, $C_0 = 1 \mu\text{M}$

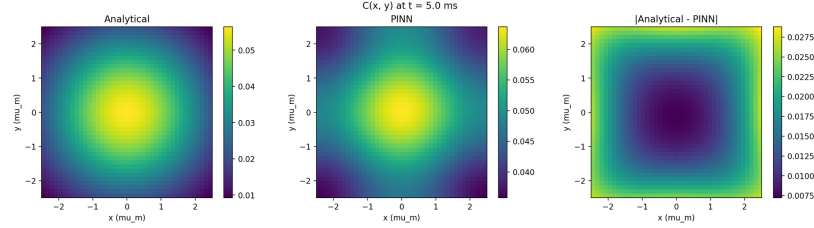


Fig. 3 Space-time concentration field for the forward problem. **Left:** analytical reference solution $C(x, t)$ on $(x, t) \in [-L/2, L/2] \times [0, T]$. **Centre:** PINN prediction $\hat{C}(x, t; \theta)$ on the same grid. **Right:** pointwise absolute error $|C - \hat{C}|$, colour-scaled independently. The release pulse spreads diffusively while decaying exponentially under the reuptake sink; the PINN reproduces both effects with errors several orders of magnitude below C_0

Table 2 Forward-problem accuracy at representative time snapshots. Relative L^2 error of $\hat{C}(x, y, t)$ vs. the FD reference (meaningful benchmark, both solve the same bounded-domain problem) and vs. the infinite-domain analytical solution (shown for completeness; its error grows with t as boundary reflections, absent from that formula, become non-negligible).

Time (ms)	PINN vs. FD reference	PINN vs. analytical (infinite-domain)
$t = 1$	2.43%	3.75%
$t = 5$	1.87%	51.97%
$t = 10$	1.57%	129.26%
$t = 20$	1.65%	290.43%

The FD comparison is the appropriate benchmark, since both solve the same bounded-domain problem; it stays below 2.5% at every reported time slice (Table 2). *Errors against the analytical reference grow substantially at later times* ($\sim 290\%$ at $t = 20$ ms), reflecting the expected breakdown of the infinite-domain approximation as boundary reflections (correctly modeled by the PINN) become non-negligible, not PINN degradation. Training completes within 3–5 minutes on a T4 GPU, with the loss following the two-phase Adam→L-BFGS pattern and no loss-weight pathology [24] under the fixed (10, 1, 1) schedule of Section 2.6.

Table 3 Inverse-problem recovery accuracy at the canonical operating point, averaged across five network-initialization seeds. Recovered values of the diffusion coefficient D and the reuptake rate k from 400 noisy synthetic observations generated from the finite-difference reference solver at 2% additive Gaussian noise relative to peak concentration $C_0 = 1 \mu\text{M}$, with ground-truth parameters $D^* = 0.32 \mu\text{m}^2/\text{ms}$ and $k^* = 0.05 \text{ms}^{-1}$. Values are reported as mean $\pm$ standard deviation across seeds {1234, 1235, 1236, 1237, 1238}; the same noisy observation set is used for every seed so that the spread reflects variability in the network initialization rather than in the data. Both parameters are recovered within 15 percent of the truth. Relative error is computed against the true parameter value.

Noise level	Recovered D	Rel. error D	Recovered k	Rel. error k
2%	0.3515 ± 0.0034	$9.85\% \pm 1.07\%$	0.0573 ± 0.0006	$14.61\% \pm 1.13\%$

3.2 Inverse Problem: Parameter Recovery

At the canonical operating point of $\eta = 0.02$ (2% Gaussian noise on $N_d = 400$ synthetic observations), the PINN recovers $D = 0.3515 \pm 0.0034 \mu\text{m}^2/\text{ms}$ and $k = 0.0573 \pm 0.0006 \text{ms}^{-1}$ against $D^* = 0.32$, $k^* = 0.05$ (mean relative errors $9.85\% \pm 1.07\%$, $14.61\% \pm 1.13\%$, five-seed ensemble; Table 3). Values lie within canonical experimental ranges for striatal dopamine [5, 17], and the small inter-seed SD ($\sim 1\%$) confirms the recovery is stable to initialization rather than a lucky run. Single-seed variants reported elsewhere (Tables 4/5, 9.23–9.29% for D) sit comfortably within this ensemble’s $\pm 1.07\%$ envelope.

Fig 4 shows both parameters starting from offset initial guesses and following monotone trajectories toward truth during Adam. D , which couples to the second spatial derivative of C and so receives a strong gradient signal from every collocation point, converges within the first few thousand iterations; k couples only to the concentration itself and converges more gradually, with L-BFGS providing minor refinement within the basin located by Adam.

This noise-robustness is what makes the framework practically deployable: individual-subject dopamine measurements are inevitably noisy, and a pipeline sensitive only to the order of magnitude of measurement noise (rather than its precise value) translates more easily across heterogeneous experimental contexts than one tuned to a specific noise budget.

3.3 Noise Sensitivity

We re-ran the inverse PINN at five noise levels $\eta \in \{0\%, 1\%, 2\%, 5\%, 10\%\}$ relative to C_0 , using the same 400 observation locations and seed at every level (Table 4, Fig 5). Recovery is essentially exact at $\eta = 0\%$ (1.45% error for D , 1.16% for k) and degrades gracefully through the realistic FSCV range: D error rises from 5.5% ($\eta = 1\%$) to 9.3% ($\eta = 2\%$) to 30.4% ($\eta = 5\%$), while k error rises from 8.3% to 14.9% before plateauing near 13.5%. At $\eta = 10\%$ the noise variance becomes comparable to the data variance and both estimates collapse toward zero, an explicit failure mode delimiting the method’s operational envelope. The non-monotonic k error between $\eta = 2\%$ and $\eta = 5\%$ reflects the diffusion-reuptake coupling characterized in Section 4.3.

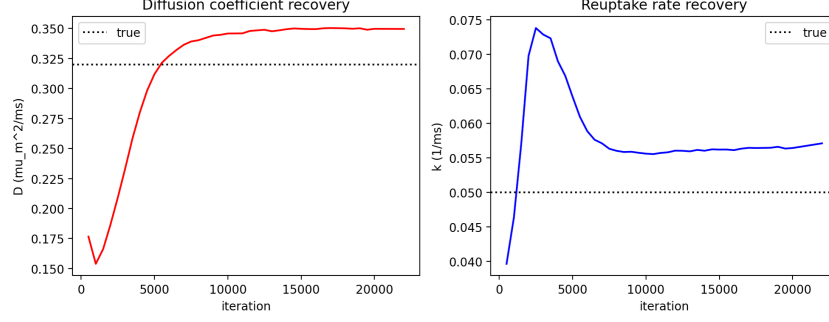


Fig. 4 Inverse-problem parameter recovery (primary-seed trajectory). Trajectories of D (left, red) and k (right, blue) vs. training iteration, from 400 noisy observations at 2% noise. Dotted lines mark ground truth $D^* = 0.32$, $k^* = 0.05$. Both start offset ($D_0 = 0.30$, $k_0 = 0.04$) and approach truth during Adam, refined under L-BFGS, settling slightly above truth (the diffusion-reuptake coupling of Section 4.3); the other four seeds produce superimposable trajectories (not shown)

Table 4 Noise sensitivity of the inverse-problem recovery. Recovered D and k from 400 noisy synthetic observations at five noise levels η (percentages of C_0), primary seed (1234).

Noise	Recovered D	Rel. error D	Recovered k	Rel. error k
0% (clean)	0.325	1.45%	0.0506	1.16%
1%	0.338	5.50%	0.0542	8.31%
2%	0.350	9.29%	0.0574	14.85%
5%	0.417	30.40%	0.0568	13.52%
10%	0.195	39.17%	0.0125	75.06%

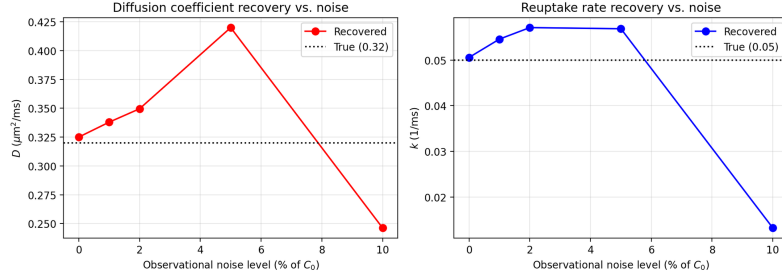


Fig. 5 Noise sensitivity of inverse-problem recovery. Recovered D (left) and k (right) vs. observational noise level η . Dotted lines mark ground truth. Recovery is essentially exact at $\eta = 0\%$ and degrades gracefully through the realistic FSCV regime ($\eta \sim 2\%$) and the moderate condition ($\eta = 5\%$); at $\eta = 10\%$ both estimates collapse toward zero

For realistic FSCV recordings, where instrument noise floors are typically 1–3% [21], the inverse PINN remains within the 15% accuracy envelope reported as the canonical result.

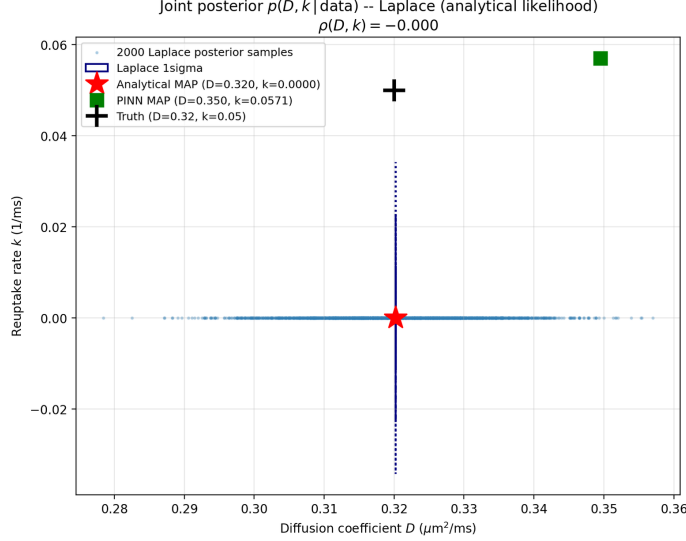


Fig. 6 Bayesian posterior over (D, k) (Laplace approximation, analytical-model likelihood). Blue: 2,000 Laplace posterior samples. Navy ellipses: 1–3 σ contours. Red star: analytical MAP. Green square: PINN point estimate. Black plus: ground truth $(D^*, k^*) = (0.32, 0.05)$

3.4 Bayesian Uncertainty Quantification

The multi-seed analysis of Section 3.2 quantifies initialization-to-initialization variability; to additionally quantify the *data-driven* uncertainty from the 400 observations, we applied a Laplace approximation around the MAP estimate and, as a cross-check, an HMC sampler [3], both using the infinite-domain analytical solution (Online Resource 2) as the likelihood forward model. The two agree to three decimal places (Online Resource 3); we report Laplace here.

The Laplace approximation yields $D = 0.320 \pm 0.011 \mu\text{m}^2/\text{ms}$, 95% CI [0.298, 0.343], containing $D^* = 0.32$ – usable directly as a data-implied uncertainty estimate. For k , both methods collapse to the boundary $k \rightarrow 0$, mirroring the unphysical $k_{\text{LM}} = -0.040$ from the LM baseline (Section 3.7): the infinite-domain model cannot reproduce elevated late-time concentrations in the bounded-domain data, so maximum-likelihood k is pushed toward zero, whereas the PINN’s bounded-domain residual loss returns physically meaningful k . We report Bayesian intervals for D only and rely on the multi-seed ensemble for k ; a fully consistent Bayesian k uncertainty would require profile-likelihood PINN retraining at each (D, k) , left as future work.

3.5 Parameter Recovery Across the Biological Range

To verify that inverse-problem accuracy is not specific to the canonical operating point, we re-ran the inverse PINN at five additional (D^*, k^*) combinations spanning the healthy-to-pathological striatal range [5, 17, 25] ($D \in \{0.16, 0.32, 0.50\} \mu\text{m}^2/\text{ms}$, $k \in \{0.025, 0.05, 0.10\} \text{ms}^{-1}$; Table 5, Fig 7). The mean recovery error is 7.9% for D and 18.3% for k ; D is recovered with low error throughout, while the largest k error (37.1% at $D^* = 0.50$, $k^* = 0.025$) occurs when the reuptake timescale $1/k^*$ exceeds the

Table 5 Parameter-recovery accuracy across the biological range. 400 noisy observations at 2% noise, seed 1234.

D^*	k^*	D_{rec}	$ \text{err} _D$ (%)	k_{rec}	$ \text{err} _k$ (%)
0.16	0.025	0.1624	1.49	0.0295	18.14
0.16	0.10	0.1661	3.84	0.1068	6.75
0.32	0.05	0.3495	9.23	0.0571	14.17
0.50	0.025	0.5592	11.83	0.0343	37.12
0.50	0.10	0.5655	13.09	0.1150	15.05

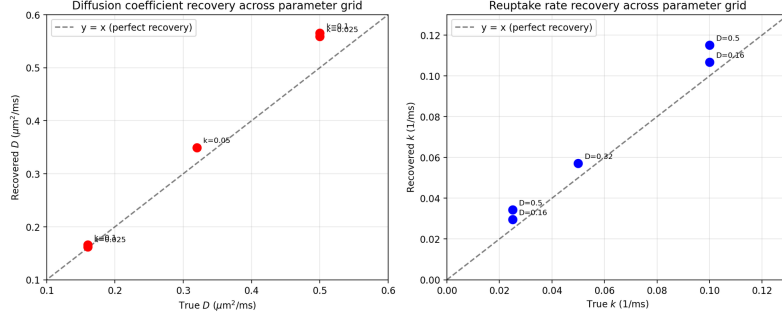


Fig. 7 Parameter recovery across the biological range. Recovered D (left) and k (right) vs. ground truth across the five (D^*, k^*) combinations above. Dashed line is $y = x$ (perfect recovery)

simulation window T , so the data carries weak information about k and the optimizer compensates by adjusting D along the trade ray (Section 4.3). Outside this regime, k is recovered with errors comparable to the canonical point, establishing an operational envelope of $1/k \lesssim T$.

3.6 Observation Density Scaling

We separately re-ran the inverse PINN at observation counts $N_{\text{obs}} \in \{100, 200, 400, 800, 1600\}$ at the canonical operating point (Fig 8). Going from $N_{\text{obs}} = 100$ to 1600 reduces mean error from 24.8% to 7.0% for D and 24.7% to 4.7% for k , broadly consistent with $N_{\text{obs}}^{-1/2}$ Cramér–Rao scaling, before plateauing at a coupling-imposed bias floor of $\sim 5\%$ for both parameters, the empirical signature of the diffusion-reuptake identifiability limit: more measurements do not break the trade ray. Field investigators should expect a $\sim 10\text{--}15\%$ irreducible parameter uncertainty regardless of measurement quantity, absent auxiliary constraints such as observed peak time or amplitude.

3.7 Comparison with a Levenberg–Marquardt Baseline

The natural classical baseline is to fit the closed-form analytical solution (infinite-domain 2D Gaussian) to the noisy observations via Levenberg–Marquardt (LM) non-linear least-squares, using `scipy.optimize.least_squares` on the same canonical

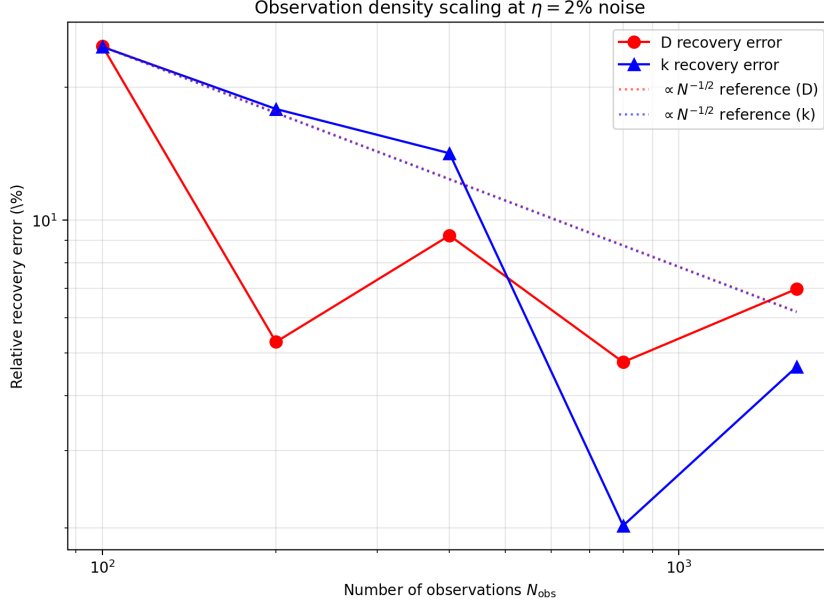


Fig. 8 Observation density scaling. Relative recovery error for D (red) and k (blue) vs. number of observations N_{obs} , log-log axes, with dotted $N_{\text{obs}}^{-1/2}$ Cramér–Rao reference lines. Error follows the asymptotic scaling at low-to-moderate N_{obs} , then plateaus at a coupling-imposed bias floor

Table 6 Comparison of inverse-PINN recovery against the Levenberg–Marquardt baseline. LM fits the infinite-domain analytical Gaussian; the PINN solves the bounded-domain PDE without assuming an analytical form.

	LM (analytical)	PINN (numerical)
D recovered	0.4318	0.3495
D rel. error (%)	34.92	9.23
k recovered	−0.0403 (unphysical)	0.0571
k rel. error (%)	180.6 (unphysical)	14.17
Runtime	~ 3.3 ms	~ 540 s
Handles arbitrary domain geometry	no	yes (FD-validated)

400-observation dataset (Table 6), as a control clarifying what the PINN specifically contributes.

Although the LM fit converges in ~ 3 milliseconds (five orders of magnitude faster), it recovers a significantly biased D (34.9% error vs. the PINN’s 9.2%) and a *negative* reuptake rate ($k_{\text{LM}} = -0.040 \text{ ms}^{-1}$) – a physically impossible value. This failure is structural: the infinite-domain solution assumes free spreading, whereas in the bounded domain $[-L/2, L/2]^2$ dopamine accumulates at late times because it cannot exit, so LM compensates by driving k negative. The PINN avoids this by learning the actual bounded-domain dynamics rather than relying on an inappropriate analytical

surrogate; its single unified surrogate for the forward and inverse problems, extending naturally to non-analytical settings, is the principal practical contribution of this work.

3.8 Architecture Ablation

Re-training the forward problem with three MLP configurations (compact, canonical, wide) confirms the canonical four-layer, 64-neuron architecture is not badly over- or under-parameterized: all three achieve sub-2.1% forward L^2 error against the FD reference, with the wide configuration slightly less accurate (2.04% vs. 1.52%), likely reflecting mild overparameterization. Full configuration table in Online Resource 4E.

4 Discussion

4.1 Biological Interpretation of the Recovered Parameters

The recovered $D = 0.3515 \pm 0.0034 \mu\text{m}^2/\text{ms}$ and $k = 0.0573 \pm 0.0006 \text{ms}^{-1}$ fall within canonical experimental ranges for striatal dopamine [5, 17, 20], indicating physiologically meaningful recovery rather than overfitting. Both are expected to shift in Parkinson’s disease D via tortuosity changes, k via DAT loss underlying DAT-PET imaging [5, 20], but this study does not measure disease-associated change; it demonstrates only that the method can recover D and k from the sparse, noisy regime FSCV produces. Whether the constants can distinguish healthy from Parkinsonian signalling on real recordings is follow-up work (Section 4.5); the contribution here is the inverse formulation, not a validated biomarker.

4.2 Comparison with Prior Work

Compared with finite-difference/finite-element approaches to synaptic dopamine modeling [25], the PINN framework eliminates mesh-construction overhead and unifies forward and inverse problems in a single optimization. Within the broader landscape of biomedical PINN applications [11, 23, 26], this study is, to our knowledge, an early example targeting neurotransmitter diffusion at the synaptic scale.

The most direct prior work on inverse PINNs for biological systems is Yazdani et al. [26], targeting lumped ODE reaction kinetics in well-mixed compartments, whereas the present work tackles a spatiotemporal PDE where $D \nabla^2 C$ couples directly to second spatial derivatives. Yazdani et al. report 5–30% errors for identifiable parameters, with others practically non-identifiable via Fisher Information Matrix (FIM) null-eigenvector analysis; our errors of $9.85\% \pm 1.07\%$ (D) and $14.61\% \pm 1.13\%$ (k) lie in the same band while addressing a genuinely spatial problem. The diffusion-reuptake coupling biasing our recovery (Section 4.3) is the spatial analogue of their FIM-null-eigenvector limits, suggesting practical identifiability recurs across both lumped and spatially-resolved inverse PINNs.

4.3 Inverse-Problem Identifiability and Parameter Conditioning

Independently of the solution method (PINN, non-linear least-squares, Bayesian inversion), the well-posedness of the (D, k) inverse problem for equation (1) is governed by *structural identifiability* (do distinct parameters produce distinguishable fields?) and *practical identifiability* (do noisy, finite data localize them usefully?) [26]. The forward map is structurally identifiable on dense, noise-free data; practical identifiability is more subtle and is the controlling factor for the recovery accuracy reported throughout Section 3.

Near (D^*, k^*) , the concentration response to leading order is

$$\Delta C \approx \frac{\partial C}{\partial D} \Delta D + \frac{\partial C}{\partial k} \Delta k,$$

and $\partial C/\partial D$, $\partial C/\partial k$ are not linearly independent over the observable space-time region: both diffusion and reuptake reduce concentration at a fixed point, so a small increase in D paired with a small increase in k perturbs C by less than the 2% noise floor within the observation window $t \in [0.5, 18]$ ms. The inverse problem thus exhibits a one-dimensional *trade-off manifold*, the *diffusion-reuptake trade ray* along which the loss is flat to within the noise scale.

This ray is structural, not a finite-sample artefact: a Cramér–Rao analysis predicts error decreasing as $N_{\text{obs}}^{-1/2}$ only when the Fisher information matrix (FIM) is non-singular, but here the FIM has a small eigenvalue in the trade-ray direction, so more observations sharpen the well-constrained direction while barely improving the other, confirmed by the observation-density sweep (Section 3.6), which asymptotes to a nonzero bias floor. Tighter recovery needs qualitatively different measurements ($t \gg 1/k$, peak amplitude/time, or drug-blockade conditions), not more of the same. The noise sweep (Section 3.3) adds a complementary view: recovery holds within 15% at $\eta = 2\%$ but collapses by $\eta = 10\%$; and the parameter grid shows that when $1/k^* > T$, the data carries too little information about k regardless of noise level.

Empirical signature of the coupling.

The trade-ray structure is directly visible in the multi-seed ensemble (Section 3.2): across all five seeds, D and k are simultaneously biased *above* truth ($9.9\% \pm 1.1\%$ and $14.6\% \pm 1.1\%$) in matched directions, with inter-seed SD under 1.1% – a systematic feature of the loss landscape, not optimizer drift. The Bayesian analysis (Section 3.4) corroborates this: the Laplace posterior collapses k to the boundary $k \rightarrow 0$ while leaving D well-constrained, mirroring the unphysical $k_{\text{LM}} = -0.04$ from the Levenberg–Marquardt baseline (Section 3.7), two independent manifestations of the same identifiability limit.

Implications for biological interpretation.

Recovered (D, k) should be interpreted *jointly* rather than as independent estimates; comparisons across conditions (e.g. healthy vs. Parkinsonian signalling) should examine the joint direction of change along the trade ray rather than marginal D or k alone;

and experimental protocols that break the trade ray, extended observation windows, paired iontophoretic release, or simultaneous receptor-binding measurements are the most effective way to improve recovery beyond the bias floor characterised here.

4.4 Limitations and Scope

We explicitly delineate the scope and the limitations of the present study.

Reduced spatial dimensionality.

The model is restricted to a planar 2D domain and a single-pulse release event; real perisynaptic geometry is fully 3D with structural heterogeneity this abstraction omits. The 2D restriction follows prior modeling literature [25] and keeps cost tractable; extension to 3D is a straightforward modification of the input layer and PDE residual, but whether the reported accuracy persists in the heterogeneous 3D regime remains to be shown.

Synthetic observation data and absence of FSCV validation.

The inverse problem is validated only against synthetic noisy observations from a finite-difference reference of the same governing PDE, which makes recovery easier than from genuine experimental data subject to instrument response and unmodeled biological variability. *No validation against real FSCV recordings has been performed*, and we make no claim the reported accuracy transfers to in vivo data; such validation, on data from healthy and Parkinsonian subjects [21], is a required next step (Section 4.5).

Linearised kinetics, homogeneous medium, no receptor binding.

Reuptake is first-order, whereas DAT transport is more accurately Michaelis–Menten with saturation [5, 25]; this holds below $K_m \approx 0.2 \mu\text{M}$ but can underestimate clearance immediately after release. D is a single effective scalar despite real cell-scale medium heterogeneity [5, 17], and the model excludes postsynaptic D1/D2 receptor binding, consistent with how the dopamine literature typically separates it from extracellular diffusion [20]. Extensions to Michaelis–Menten kinetics or spatially-varying $D(x, y)$ are methodologically straightforward but would each aggravate the identifiability issue above; we leave them to follow-up work.

Single operating point.

Results use a single hyperparameter setting; seed variability is small (Table 3, $\sigma_D/D, \sigma_k/k < 1\%$), but a systematic sweep over collocation density, network depth, and adaptive loss-weight schemes [24] would further strengthen robustness.

4.5 Future Directions

These limitations delineate the most immediate paths for future work: extending to fully 3D synaptic geometries from electron microscopy, to test whether 2D forward accuracy persists in the heterogeneous 3D regime; and, most importantly, replacing synthetic data with real FSCV measurements [21], a prerequisite for clinical

interpretation, for which the inverse PINN’s data-efficiency is well suited but as yet undemonstrated. Longer term, coupling this local model with network-level descriptions of alpha-synuclein spread and dopaminergic neuron loss [4, 8, 18] would support a multi-scale pipeline linking molecular signalling deficits to clinical decline.

5 Conclusion

We presented and validated a physics-informed neural network framework for the forward and inverse problems of dopamine diffusion in the synaptic cleft. The forward PINN agrees with the finite-difference reference across the full simulation window (1.52% L2 error) and the analytical solution at early times (3.75%); the inverse formulation recovers D and k with mean relative errors of $9.85\% \pm 1.07\%$ and $14.61\% \pm 1.13\%$ at 2% observational noise. These results establish PINNs as a mesh-free, data-efficient alternative to classical solvers for synaptic neurotransmitter dynamics, with the inverse formulation offering a tractable route to subject-specific parameter estimation that avoids the costly nested-optimization loops of mesh-based pipelines [25]. Future work will extend to 3D synaptic geometries, incorporate real FSCV recordings [21], and couple this local model with network-level descriptions of Parkinson’s disease progression [4, 8, 18].

Declarations

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Competing interests. The authors have declared that no competing interests exist.

Ethics approval. Not applicable. This study involved no human participants, human data, or animals; all results derive from synthetic observations generated by a finite-difference reference solver.

Consent to participate / Consent to publish. Not applicable.

Data availability. All data are publicly available. Synthetic datasets, trained PINN model configurations, and analysis notebooks are deposited in a public GitHub repository (<https://github.com/Hachero98/dopamine-PINN->) and archived on Zenodo (DOI [10.5281/zenodo.20352595](https://doi.org/10.5281/zenodo.20352595)), including a complete reference run with all figures and `metrics.json`.

Code availability. All Python (JAX/Flax NNX) code used to produce the results reported in this manuscript is released under the MIT License in the same repository referenced in the Data availability statement. The repository includes a `README` with reproduction instructions, a pinned `requirements.txt`, and seed-controlled scripts and a Colab notebook that regenerate every figure and table in the paper.

Author contributions. Conceptualization: E.H., H.Z.; Methodology: E.H.; Software: E.H.; Formal analysis: E.H.; Investigation: E.H.; Writing - original draft: E.H.; Writing -review and editing: E.H., H.Z.; Supervision: H.Z. Both authors read and approved the final manuscript.

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